PREMIER UNIVERSITY, CHATTOGRAM

Department of Computer Science And Engineering



Database Management System Laboratory Project Report

On

Research Collaboration Platform

SUBMITTED BY

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29 September, 2026

Project Name : Cholo Research Kori

Abstract :

Cholo Research Kori is a university research collaboration platform designed to connect students, researchers, and faculty members in a centralized environment. The system allows users to discover research opportunities, create and manage research projects, collaborate with other researchers, and track research activities. The main goal of the project is to make university research collaboration more organized, accessible, and efficient.

Introduction :

Research plays an important role in university education and innovation. However, students often face difficulties in finding suitable research opportunities, researchers, or faculty members working in their areas of interest. Research activities are also frequently managed through different communication platforms, making collaboration and project management difficult. To solve these problems, we propose "Cholo Research Kori", a web-based research collaboration platform for university students, researchers, and faculty members. The platform provides users with profiles containing their academic information, research interests, skills, and previous research activities. Users can discover research projects, researchers, and available collaboration opportunities through the system. The system also provides research project management features where supervisors can create research projects and labs, manage project members, assign tasks, define milestones, and track research progress. Students and researchers can request to join projects and communicate with other collaborators. The system also supports publication management and basic research analytics, creating a complete environment for managing university research collaboration.

Objective :

The main objectives of the Cholo Research Kori project are:

- To create a centralized platform for university research collaboration.
- To allow students and researchers to discover suitable research projects and research opportunities.

- To connect students with researchers and faculty members according to their research interests and skills.
- To allow supervisors to create and manage research projects and research labs.
- To provide a structured collaboration request system.
- To manage project members, tasks, and research milestones.
- To manage research publications and research outputs.
- To make university research activities more organized and accessible.

The absence of a centralized research collaboration platform creates communication gaps and makes research management less organized. Cholo Research Kori addresses this problem by connecting research discovery, collaboration, project management, communication, and publication tracking within one system.

Methodologies :

The proposed system is divided into several panels and functional modules with independent responsibilities :

User / Researcher Panel : Handles registration, login, profile management, research interests, skill information, project searching, collaboration requests, task tracking, and publication access.

Supervisor / Faculty Panel : Handles project creation, lab management, project supervision, collaborator review, request approval, task assignment, milestone management, and research progress monitoring.

Research Project Panel : Storers research project data, project members, project status, tasks, deadlines, milestones, and publication links.

Collaboration Panel : Managing sending, receiving, reviewing, and accepting collaboration requests and joining requests between users and projects.

Messaging Panel : Supports conversations, message exchange, participant management, and communication between project members.

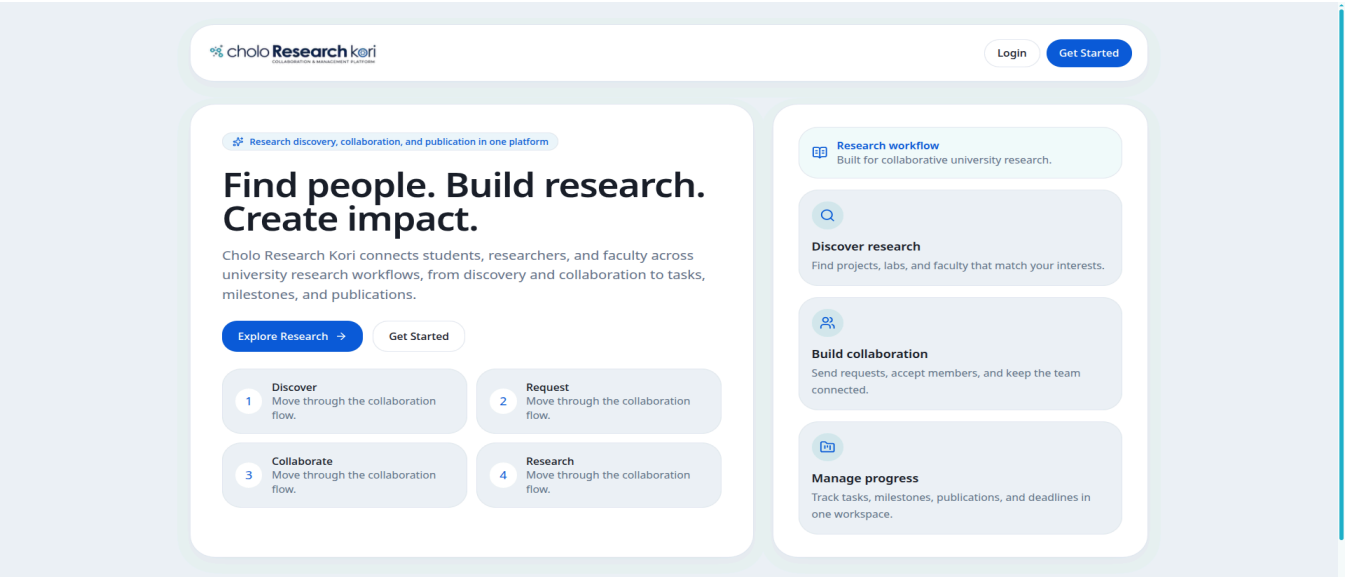
Notification Panel : Displays system notifications for project updates, collaboration requests, and message-related activities, while separating read and unread notifications.

Profile / Settings Panel : Manages user profile information, academic details, research interests, skills, biography, and account preferences.

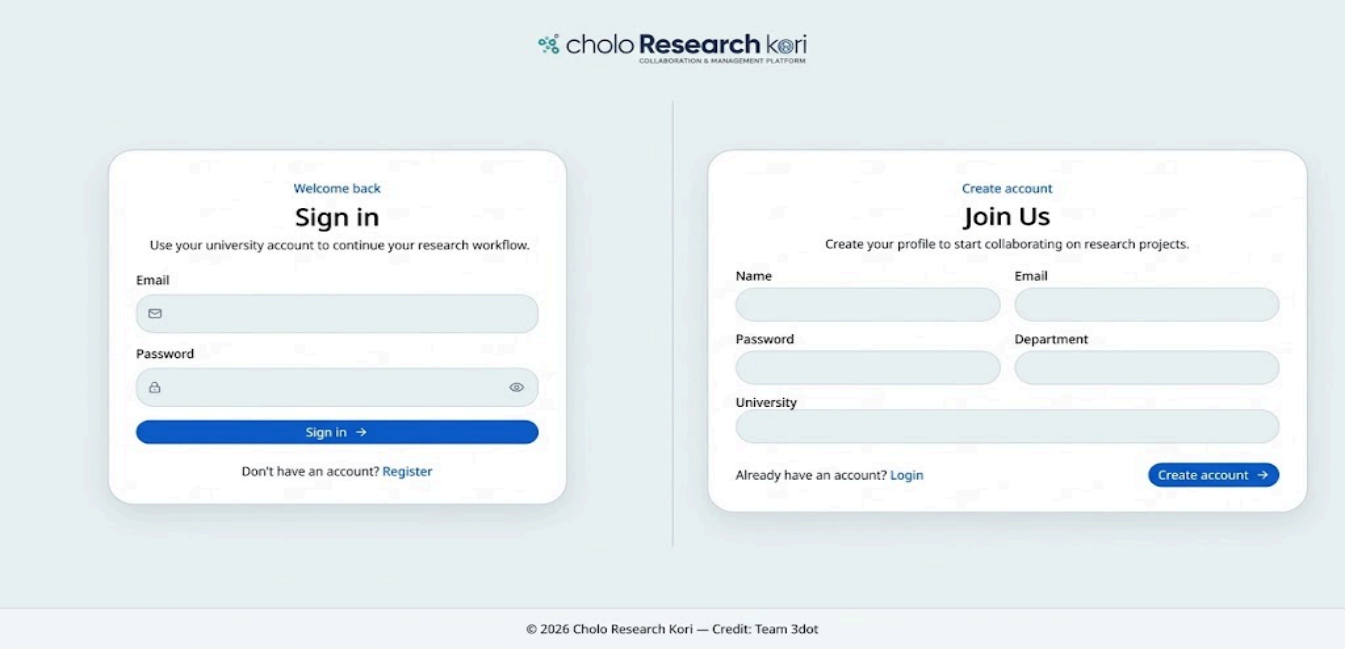
Database Management : Stores all system information in a relational database using primary and foreign keys, maintaining entity relationships and reducing data redundancy through normalization.

System Outputs : The following screenshots demonstrate the major interfaces and functional features of the Cholo Research Kori system. The screenshots present the complete user journey from accessing the public landing page to authentication, dashboard interaction, research project management, communication, collaboration requests, notifications, and profile settings.

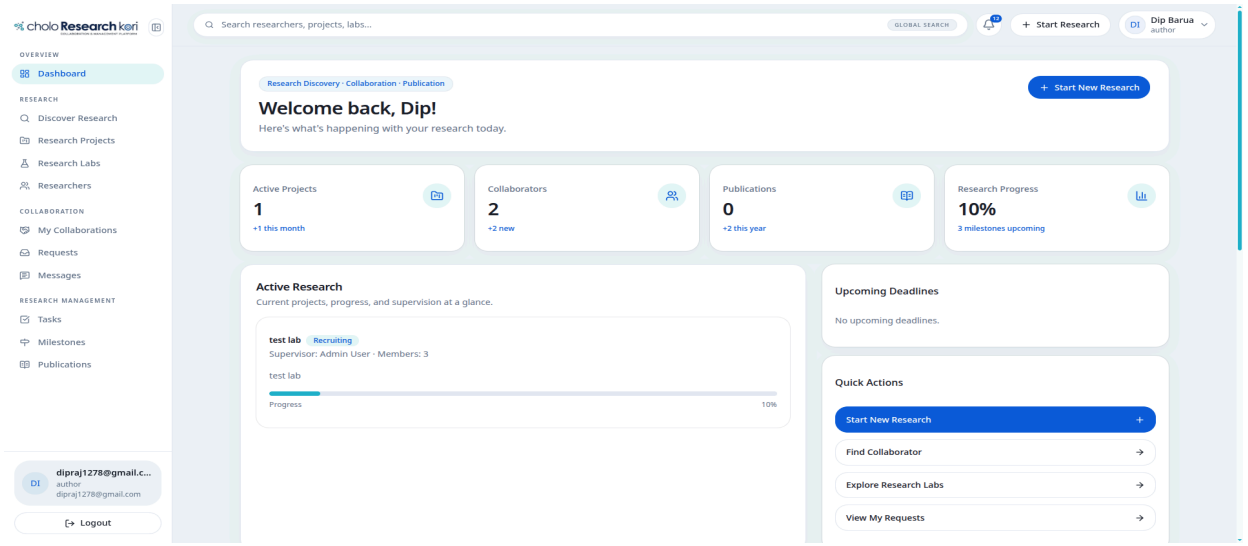
LANDING / HOME PAGE :



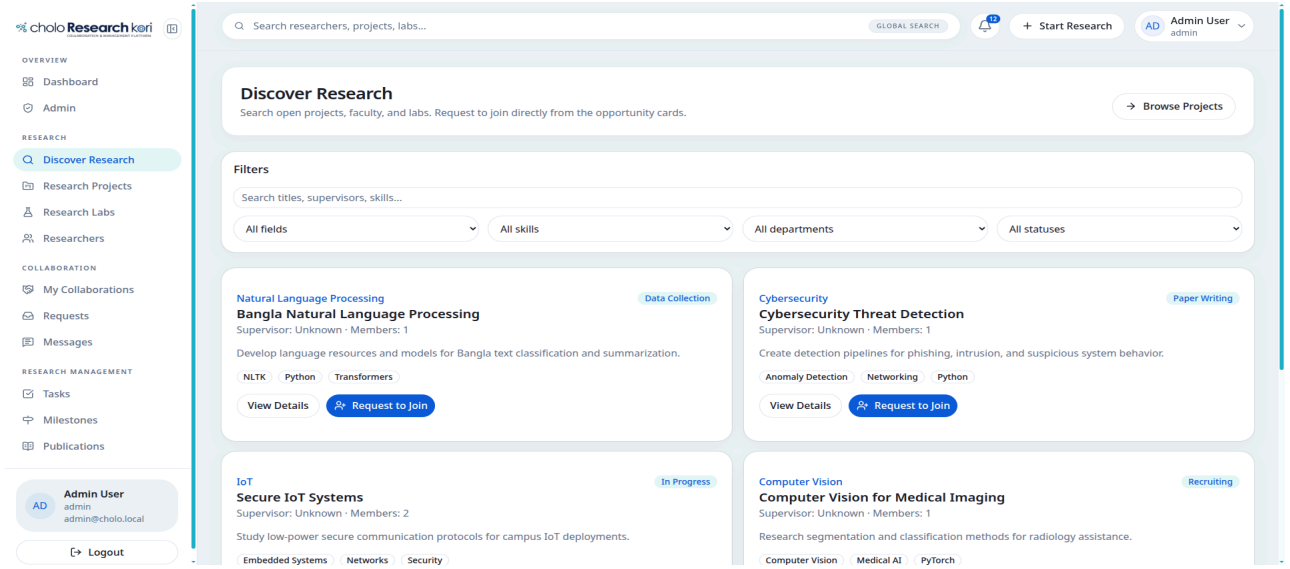
Login and Signup :



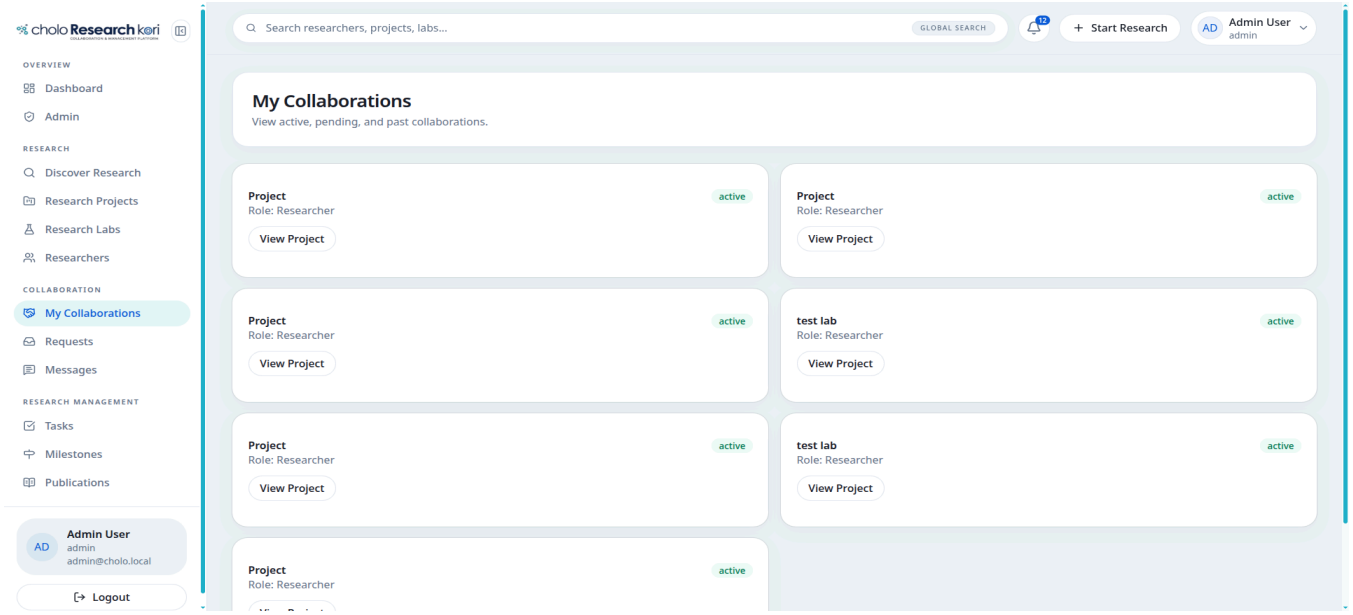
Dashboard :



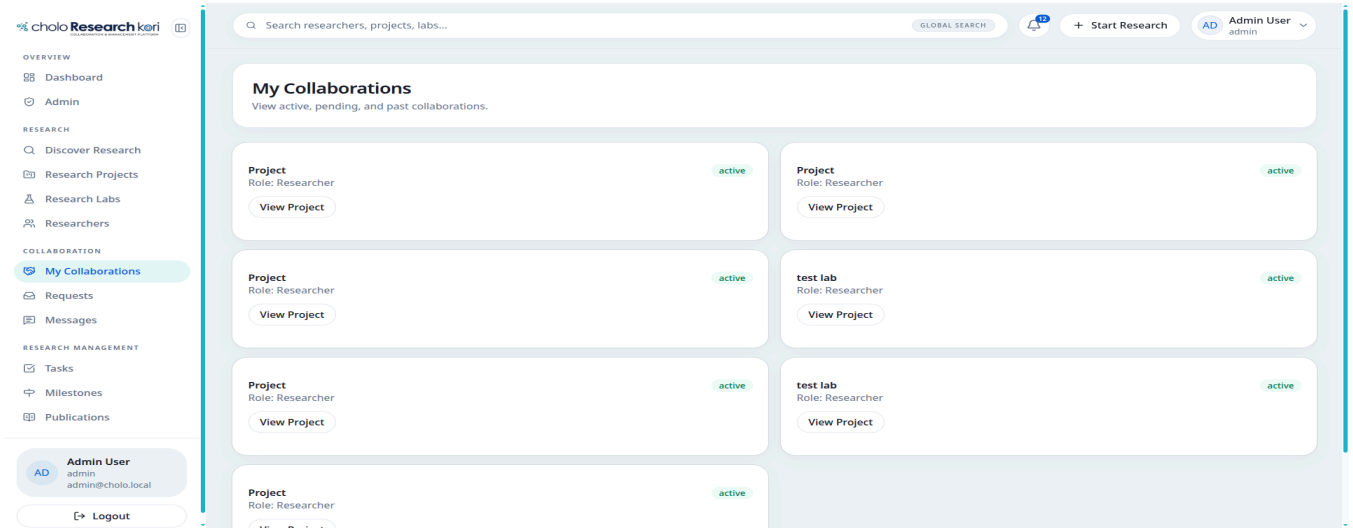
Discover Research :



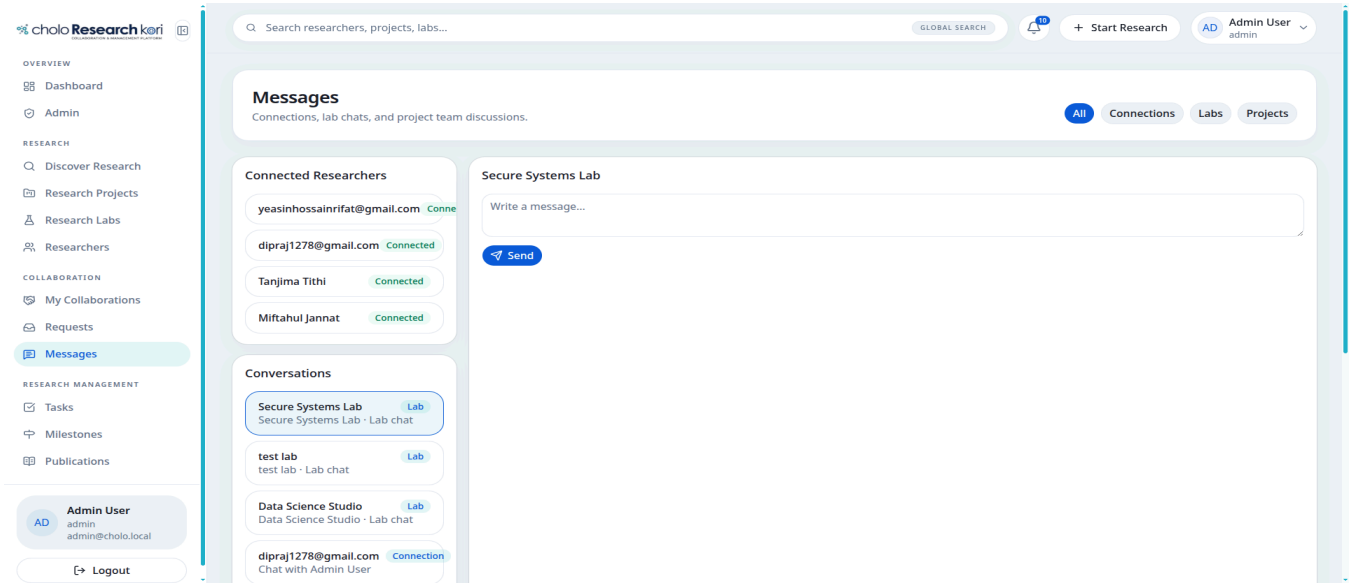
My Collaborations :



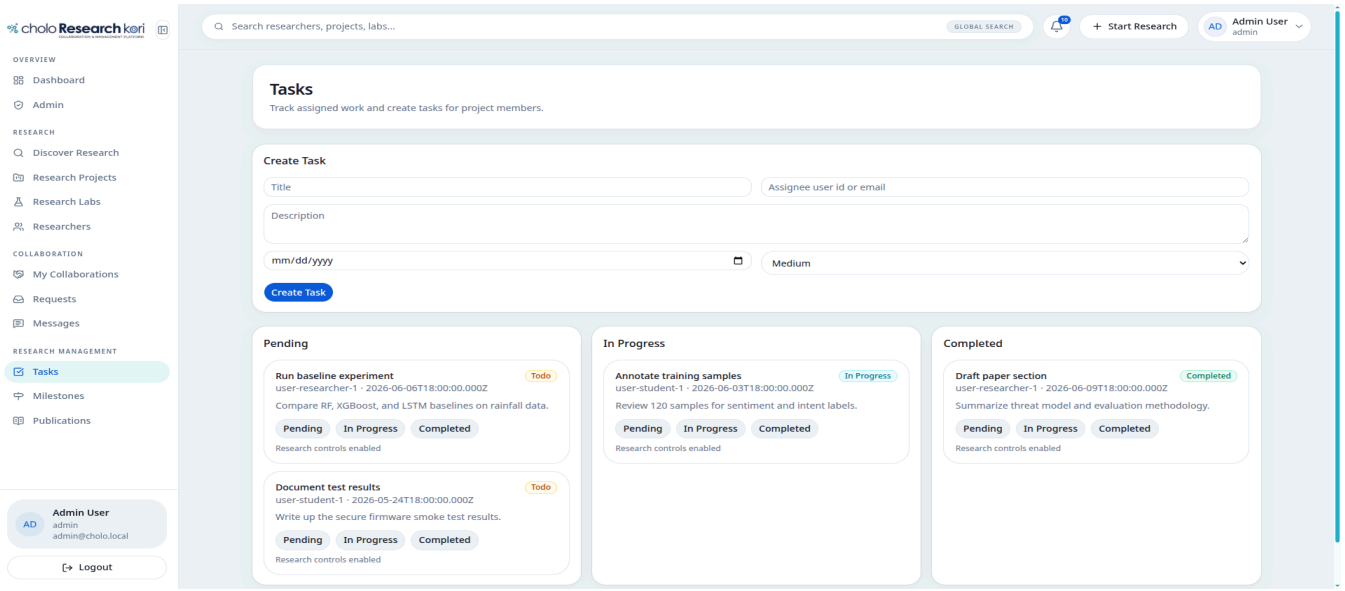
Collaboration Requests :



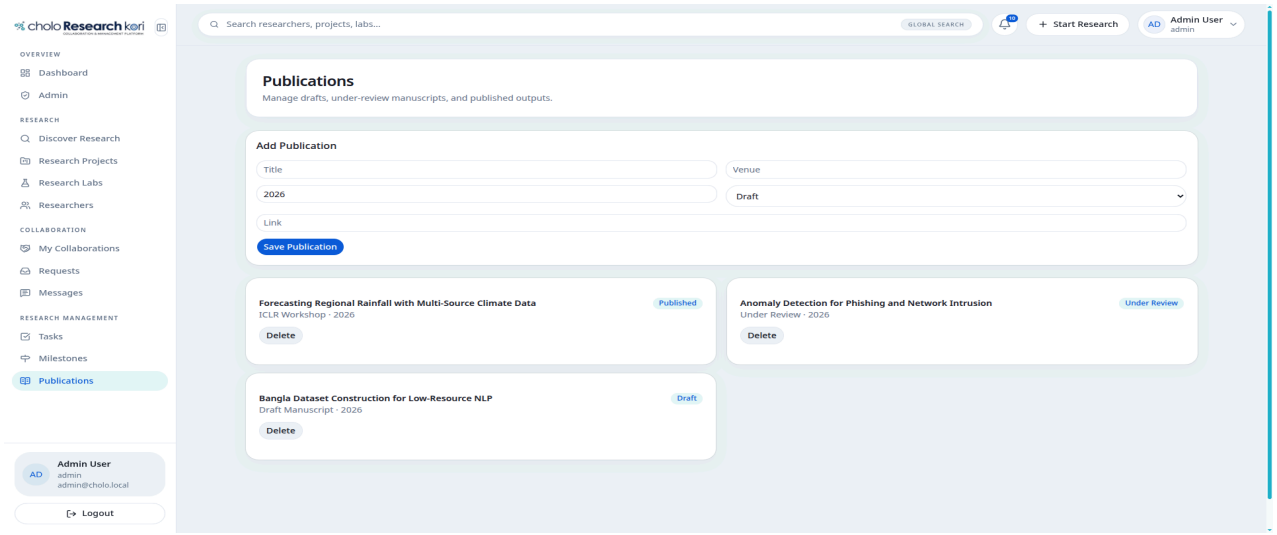
Inbox Messages :



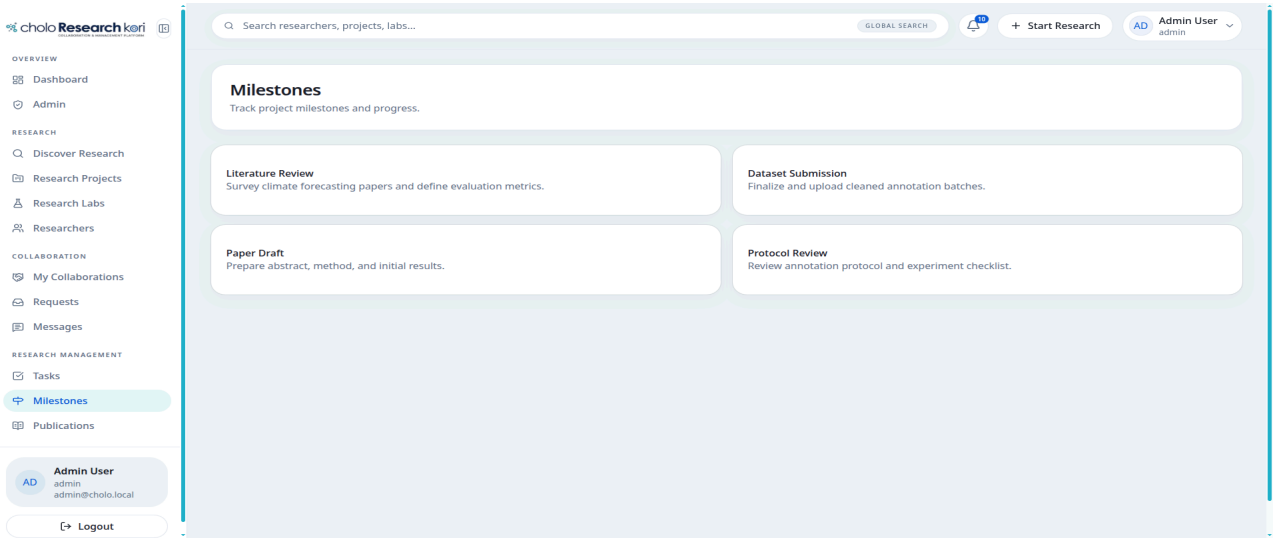
Tasks :



Publications :



Milestones :



Entities and Relationships :

Entity A	Relationship	Entity B	Meaning
USERS	supervises (1:N)	PROJECTS	One supervisor can supervise many projects.
USERS	joins (M:N)	PROJECTS (via PROJECT_MEMBERS)	Many users can join many projects.
USERS	supervises (1:N)	LABS	One supervisor can supervise many labs.
USERS	belongs to (M:N)	LABS (via LAB_MEMBERS)	Many users can belong to many labs.
LABS	hosts/belongs (M:N)	PROJECTS (via LAB_PROJECTS)	A lab can host many projects and a project can belong to many labs.
USERS	joins (M:N)	CONVERSATIONS (via CONVERSATION_PARTICIPANTS)	Many users can join many conversations.
CONVERSATIONS	contains (1:N)	MESSAGES	One conversation has many messages.
PROJECTS	has (1:N)	TASKS / MILESTONES	One project has many tasks and milestones.
USERS	writes (M:N)	PUBLICATIONS (via PUBLICATION_AUTHORS)	Many authors can write many publications.

Table 1. Entities and relationships in the E-R model.

Database Table :

USERS

Key	Attribute	Data type	References
PK	id	VARCHAR(36)	-
	name	VARCHAR(100)	-
	email	VARCHAR(100)	-
	password	VARCHAR(255)	-
	university	VARCHAR(100)	-
	department	VARCHAR(100)	-
	role	VARCHAR(50)	-
	bio	TEXT	-
	research_interests	JSON	-
	skills	JSON	-
	avatar_seed	VARCHAR(50)	-
	joined_at	TIMESTAMP	-
	active	BOOLEAN	-

PROJECTS

Key	Attribute	Data type	References
PK	id	VARCHAR(36)	-
FK	supervisor_id	VARCHAR(36)	USERS(id)
	title	VARCHAR(100)	-
	description	TEXT	-
	field	VARCHAR(50)	-
	skills	JSON	-
	status	VARCHAR(20)	-
	progress	INT	-
	max_members	INT	-
	start_date	DATE	-
	end_date	DATE	-
	created_at	TIMESTAMP	-
	updated_at	TIMESTAMP	-

LABS

Key	Attribute	Data type	References
PK	id	VARCHAR(36)	-
FK	supervisor_id	VARCHAR(36)	USERS(id)
	name	VARCHAR(100)	-
	department	VARCHAR(100)	-
	description	TEXT	-
	research_areas	JSON	-
	created_at	TIMESTAMP	-
	updated_at	TIMESTAMP	-

PROJECT_MEMBERS

Key	Attribute	Data type	References
PK	id	VARCHAR(36)	-
FK	project_id	VARCHAR(36)	PROJECTS(id)
FK	user_id	VARCHAR(36)	USERS(id)
	joined_at	TIMESTAMP	-

LAB_MEMBERS

Key	Attribute	Data type	References
PK	id	VARCHAR(36)	-
FK	lab_id	VARCHAR(36)	LABS(id)
FK	user_id	VARCHAR(36)	USERS(id)
	joined_at	TIMESTAMP	-

LAB_PROJECTS

Key	Attribute	Data type	References
PK	id	VARCHAR(36)	-
FK	lab_id	VARCHAR(36)	LABS(id)
FK	project_id	VARCHAR(36)	PROJECTS(id)
	created_at	TIMESTAMP	-

COLLABORATION_REQUESTS

Key	Attribute	Data type	References
PK	id	VARCHAR(36)	-
FK	sender_id	VARCHAR(36)	USERS(id)
FK	receiver_id	VARCHAR(36)	USERS(id)
FK	project_id	VARCHAR(36)	PROJECTS(id)
FK	lab_id	VARCHAR(36)	LABS(id)
	message	TEXT	-
	status	VARCHAR(20)	-
	created_at	TIMESTAMP	-
	updated_at	TIMESTAMP	-

COLLABORATIONS

Key	Attribute	Data type	References
PK	id	VARCHAR(36)	-
FK	project_id	VARCHAR(36)	PROJECTS(id)
FK	lab_id	VARCHAR(36)	LABS(id)
	user_ids	JSON	-
	role	VARCHAR(50)	-
	status	VARCHAR(20)	-
	started_at	TIMESTAMP	-
	created_at	TIMESTAMP	-
	updated_at	TIMESTAMP	-

COLLABORATION_MEMBERS

Key	Attribute	Data type	References
PK	id	VARCHAR(36)	-
FK	collaboration_id	VARCHAR(36)	COLLABORATIONS(id)
FK	user_id	VARCHAR(36)	USERS(id)
	role	VARCHAR(50)	-
	created_at	TIMESTAMP	-

CONVERSATIONS

Key	Attribute	Data type	References
PK	id	VARCHAR(36)	-
FK	project_id	VARCHAR(36)	PROJECTS(id)
FK	lab_id	VARCHAR(36)	LABS(id)
	subject	VARCHAR(200)	-
	participant_ids	JSON	-
	message_ids	JSON	-
	conversation_type	VARCHAR(50)	-
	created_by	VARCHAR(36)	USERS(id)
	created_at	TIMESTAMP	-
	updated_at	TIMESTAMP	-

MESSAGES

Key	Attribute	Data type	References
PK	id	VARCHAR(36)	-
FK	conversation_id	VARCHAR(36)	CONVERSATIONS(id)
FK	sender_id	VARCHAR(36)	USERS(id)
FK	receiver_id	VARCHAR(36)	USERS(id)
	content	TEXT	-
	read_flag	BOOLEAN	-
	created_at	TIMESTAMP	-

MILESTONES

Key	Attribute	Data type	References
PK	id	VARCHAR(36)	-
FK	project_id	VARCHAR(36)	PROJECTS(id)
	title	VARCHAR(100)	-
	description	TEXT	-
	progress	INT	-
	due_date	DATE	-
	status	VARCHAR(20)	-
	created_by	VARCHAR(36)	USERS(id)
	created_at	TIMESTAMP	-
	updated_at	TIMESTAMP	-

NOTIFICATIONS

Key	Attribute	Data type	References
PK	id	VARCHAR(36)	-
FK	user_id	VARCHAR(36)	USERS(id)
	message	TEXT	-
	type	VARCHAR(50)	-
	is_read	BOOLEAN	-
	created_at	TIMESTAMP	-

TASKS

Key	Attribute	Data type	References
PK	id	VARCHAR(36)	-
FK	project_id	VARCHAR(36)	PROJECTS(id)
FK	assigned_to	VARCHAR(36)	USERS(id)
FK	created_by	VARCHAR(36)	USERS(id)
	title	VARCHAR(100)	-
	description	TEXT	-
	priority	VARCHAR(20)	-
	status	VARCHAR(20)	-
	due_date	DATE	-
	created_at	TIMESTAMP	-
	updated_at	TIMESTAMP	-

CONVERSATION_PARTICIPANTS

Key	Attribute	Data type	References
PK	id	VARCHAR(36)	-
FK	conversation_id	VARCHAR(36)	CONVERSATIONS(id)
FK	user_id	VARCHAR(36)	USERS(id)
	joined_at	TIMESTAMP	-

Fig : Database Table for Research sharing project

Relational Entities :

USERS(id PK, name, email, password, university, department, role, bio, research_interests, skills, avatar_seed, joined_at, active)

PROJECTS(id PK, supervisor_id FK, title, description, field, skills, status, progress, max_members, start_date, end_date, created_at, lab_id FK, updated_at)

PROJECT_MEMBERS(id PK, project_id FK, user_id FK, joined_at)

PROJECT_SKILLS(id PK, project_id FK, skill)

LABS(id PK, supervisor_id FK, name, department, description, research_areas, created_at, updated_at)

LAB_MEMBERS(id PK, lab_id FK, user_id FK, joined_at)

LAB_PROJECTS(id PK, lab_id FK, project_id FK, created_at)

COLLABORATION_REQUESTS(id PK, sender_id FK, receiver_id FK, project_id FK, lab_id FK, message, status, created_at, updated_at)

COLLABORATIONS(id PK, project_id FK, lab_id FK, user_ids, role, status, started_at, created_at, updated_at)

COLLABORATION_MEMBERS(id PK, collaboration_id FK, user_id FK, role, joined_at)

CONVERSATIONS(id PK, subject, participant_ids, message_ids, conversation_type, project_id FK, lab_id FK, created_at, updated_at)

CONVERSATION_PARTICIPANTS(id PK, conversation_id FK, user_id FK, joined_at)

MESSAGES(id PK, conversation_id FK, sender_id FK, receiver_id FK, content, read_flag, created_at)

TASKS(id PK, project_id FK, assigned_to FK, created_by FK, title, description, priority, status, due_date, created_at, updated_at)

MILESTONES(id PK, project_id FK, title, description, due_date, status, progress, created_by FK, created_at, updated_at)

PUBLICATIONS(id PK, project_id FK, author_ids, title, venue, year, status, link, created_by FK, created_at, updated_at)

PUBLICATION_AUTHORS(id PK, publication_id FK, author_id FK, author_order)

NOTIFICATIONS(id PK, user_id FK, type, title, message, href, related_id, read_flag, created_at, updated_at)

SETTINGS(id PK, user_id FK, notifications, appearance, privacy, created_at, updated_at)

AUTH_SESSIONS(id PK, user_id FK, created_at, updated_at)

APP_STATE(id PK, payload, updated_at)

ER Schema :

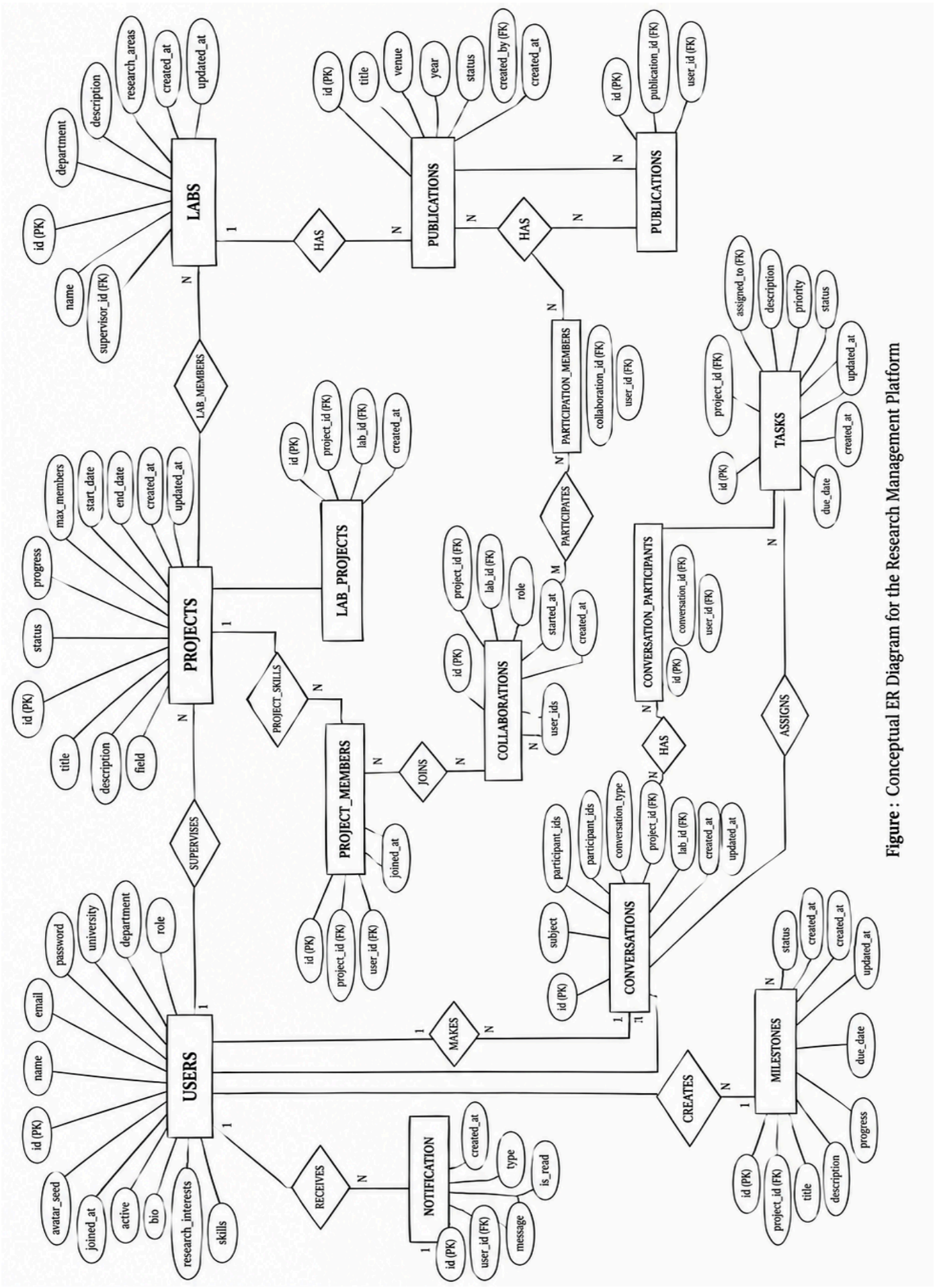


Figure : Conceptual ER Diagram for the Research Management Platform

Normalization :

1NF :

- Each table stores atomic values and each row has a unique identifier.
- Repeating values such as project members, skills, authors, and conversation participants are moved into separate tables instead of being stored as repeated columns.

2NF :

- Most tables use a single primary key (id), so there is no partial dependency problem.
- Many-to-many relationships are handled by link tables like PROJECT_MEMBERS, LAB_MEMBERS, and PUBLICATION_AUTHORS.

3NF :

- Non-key attributes depend directly on the main entity, not on other non-key attributes.
- The core design is close to 3NF because the schema separates entities and association tables instead of storing redundant data in one table.

BCNF :

- The design is also close to Boyce-Codd Normal Form because each determinant is a candidate key .
- Independent entities and link tables prevent non-key dependencies that would violate BCNF

Conclusion :

This database is designed for a research collaboration platform where users, projects, labs, communication, and publication workflows are all connected. The design uses separate entity tables and join tables to keep data organized, reduce redundancy, and support many-to-many relationships efficiently. Overall, the schema is suitable for a DBMS assignment and reflects the actual implementation of this project.